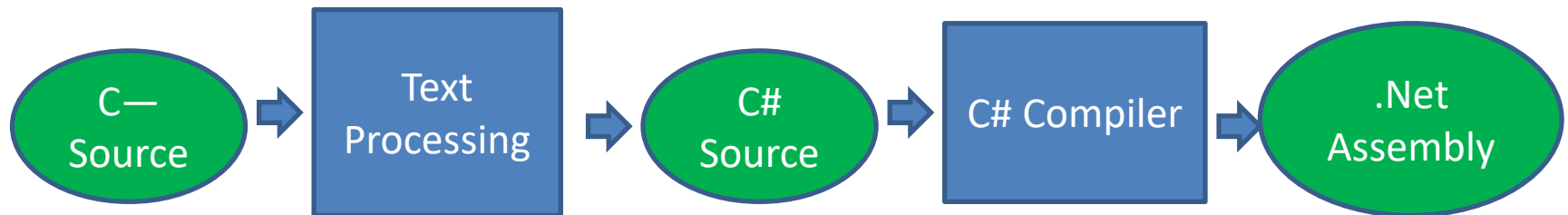


C-- Programming

What is C--



What is C--

- C-- is a subset of C#
- C-- is C# without OOP (Object Oriented Programming)
- C-- is a Procedure Language (like C Language)
- C-- is a Strongly typed Language
- C-- is a compiled Language

Gramma of C--

- Just Like C#
- C-- = C Language - Point + 2D dynamic Array

Data Type of C--

- byte: unsigned 8 bits integer
- int: 32 bits integer
- float: 32 bits real
- double: 64 bits real
- string:

Array

- 1 D array: `data_type []` // `[]`:square brackets
- 2D array: `data_type [,]` // `,`comma
- Create Array
 - 1D:
`data_type [] array_name = new data_type[len];`
 - 2D:
`datatype[,] array_name = new datatype[len0,len1];`

Array

- Get Array Size
 - 1D: `a.Length` or `a.GetLength(0)`
 - 2D: `a.GetLength(0)` ,`a.GetLength(1)`

Operators

- Just like C/C++,C#, Java
- Mathematical: +,-,*,/,+ =,- =,* =,/ =,
- Logical: &&, ||, !
- Shift:<<,>>
- Bits Operator:& ,|
- Relational: >,>=,<,<=,==,!=

Statements

Category	Examples
Declaration	<code>int a;</code> <code>byte[,] b;</code>
Expression	<code>a = 10;</code> <code>B = new byte[100,100];</code>
Selection	<code>if (f[x,y]>128) g[x,y] = 255;</code> <code>else g[x,y]=0;</code>
Iteration(loop)	<code>for (int y=0;y<h;y++)</code> <code>{</code> <code>}</code> While, do / while
Jump Statements	<code>goto, break, continue ,return</code>

Image Data Type

- byte: Gray Level Image
- ARGB: Color Image

```
struct ARGB
```

```
{
```

```
Byte A;
```

```
Byte R;
```

```
Byte G;
```

```
Byte B;
```

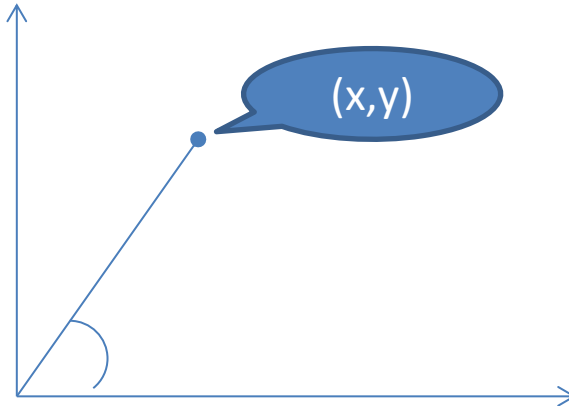
```
}
```

Image Functions

- `Byte[,] LoadImg();`
- `ARGB[,] LoadColorImg();`
- `Void ShowImg(string title, byte[,]f);`
- `Void ShowImg(string title, ARGB[,]f);`

Mathematical Functions

- `double Sin(double a);`
- `double Cos(double a);`
- `double Atan2(double y ,double x);`
- Trigonometric functions use radian mode for angles.



Mathematical Functions

<code>double Sqrt(double x);</code>	$\sqrt{x}$
<code>double Pow(double x,double y);</code>	x^y
<code>double Exp(double x);</code>	e^x
<code>double Rand();</code>	Return random value [0..1]